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- This problem set is intended for screening PhD applicants. If you find the level too challenging, or if these topics do not interest you, I do not recommend applying to the PhD programme under my supervision.
- You are not expected to complete all problems. You may choose a subset of projects that interest you and work on them as thoroughly as you can. This is not an exam; you may use any references, including AI tools.
- I am not looking only for written solutions. I care more about your reasoning.
- Part of the qualifying exam for current PhD students at the end of Year 1 is based on this portfolio.
- There may be a few typos, or even mistakes, in the problems. You should try to identify and correct them on your own.
- The order of the problems does not reflect their difficulty.

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Numerical PDE：数值偏微分方程

传统数值分析到机器学习求解PDE

1 Project: Numerical PDE

Consider the following one dimensional the boundary value problem (BVP) for u:

$$-u''(x) + \pi^2 \cos(\pi x) u(x) = f(x), \quad x \in [0,1] \tag{1}$$

with boundary conditions:

$$\begin{aligned} u(0) &= 0, \\ u(1) &= 0. \end{aligned} \tag{2}$$

1.有限差分法

1. Assume the function $u(x) = \sinh(\sin(\pi x))$ is the solution. Write the expression of $f(x)$. And then use this f to solve the following problems with u unknown.

2. We want to solve this BVP by finite difference method. We begin by discretizing the interval $[0,1]$. For this, consider the uniform gridpoints:

$$x_i = ih, \quad i = 0, 1, \ldots, n+1, \quad h = \frac{1}{n+1}. \tag{3}$$

Now we approximate the second derivative of a function g. Show that if g has four continuous derivatives, then

$$\frac{g_{i+1} - 2g_i + g_{i-1}}{h^2} = g_i'' + O(h^2), \tag{4}$$

where $g_i = g(x_i)$.

3. Consider now the linear system of equations

$$-\frac{g_{i+1} - 2g_i + g_{i-1}}{h^2} + \pi^2 \cos^2(\pi x_i)\, g_i = f(x_i) \quad i = 1, 2, \ldots, n. \tag{5}$$

Show that this can be rewritten in matrix form as

$$\boldsymbol{A}\boldsymbol{g} = \boldsymbol{f},$$

where $\boldsymbol{g} = (g_1, \ldots, g_n)^T$, $\boldsymbol{f} = (f_1, \ldots, f_n)^T$, and the matrix $\boldsymbol{A}$ is tridiagonal. Write the expression of the matrix $\boldsymbol{A}$.

4. Show that the finite difference scheme (5) is second-order accurate.

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5. Solve the system of equations (5) by a computer. Use the following values: $n = 10, 20, 40, 80, 160, 320$. For each $h = 1/(n+1)$, compute the error

$$e(h) = \sup_{1 \le i \le n} |g_i - u(x_i)| \tag{6}$$

and show a log-log plot of $e(h)$, that is, plot $\log(e(h))$ as a function of $\log(h)$. Verify $e(h) = O(h^2)$.

2. 神经网络求解PDE

6. Consider a (one-hidden layer) neural network approximation in the following parametric form

$$u_{\mathrm{NN}}(x) = x(1-x)\left(\sum_{i=1}^{n} a_i \rho(w_i x + b_i)\right), \tag{7}$$

which satisfies the boundary condition (2). n is the number of neurons. ρ is an activation function. To find an approximate solution of the PDE, one can minimize the loss function in the least-squares sense over the unknown parameters (a, w, b): Loss Function :

$$\min_{a,b,w} J[a,w,b] := \int_0^1 \left(-u''_{\mathrm{NN}}(x) + \pi^2 \cos^2(\pi x) u_{\mathrm{NN}}(x) - f(x)\right)^2 dx. \tag{8}$$

Find the first-order optimality condition for this minimization problem and is this a convex problem ?

6. Choose $\rho(x) = \sin(x)$ (or other functions like $\tanh$, $\max(x^3, 0)$). Use the Monte Carlo method to approximate the loss function (7) with M uniform samples $\{x_j\}_{j=1}^{M}$. Use stochastic gradient descent method (or other accelerated method like Adam) to find an optimal set of parameters $\{a_i, w_i, b_i\}_{i=1}^{n}$, by testing different $n, M = 10, 20, 40, 80, 160, 320$. What initial guess would you apply? For each n, compute the max norm error

$$e(n, M) = \|u_{\mathrm{NN}} - u\|_\infty \tag{9}$$

Build a table of $e(n, M)$ and discuss empirically how n and M affect the error.

7. What if you use the following approximate scheme

$$u_{\mathrm{NN}}(x) = x(1-x)\frac{1}{n}\left(\sum_{i=1}^{n-1} a_i \rho(w_i x + b_i)\right) \tag{10}$$

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while using all the same initials before?

3. 神经算子

8. Now let the forcing term f is arbitrary, and the neural operator (Journal of Machine Learning Research 24 (2023) 1-97) is to learn the map from f to u. First, choose a non-local integral kernel operator $\mathcal{K}$ (such as a Fourier operator) to construct a linear functional $f(x) \mapsto w_i f(x) + \int \mathcal{K}(x,y)(y)dy + b_i$, then

$$u(x) \approx \sum_i a_i \rho(w_i f(x) + \int \mathcal{K}(x,y)(y)dy + b_i(x))$$

where ρ is the activation as before. How would you learn the parameters $\{a_i, w_i, b_i\}_{i=1}^n$ and the kernel $\mathcal{K}$? How would you sample f? How do you validate your numerical result is true? How to improve the accuracy ? You may implement a simple version of this algorithm.

物理信息神经网络的失效

2 Failure of the PINN

PINN (physics-informed neural network) may fail for time-dependent solutions. Let $p(x,t)$ be the true classic solution of the Fokker–Planck equation on a torus (periodic boundary)

$$\frac{\partial p(x,t)}{\partial t} = \mathcal{L}p(x,t), \tag{11}$$

and assume p_∞ is the stationary solution that $\mathcal{L}p_\infty = 0$.

Define the sequence of the functions

$$f_k = \begin{cases} p(x,t), & t \le \frac{1}{k}, \\ \phi_k(t)p(x,t) + (1-\phi_k(t))p_\infty(x), & t > \frac{1}{k}, \end{cases} \tag{12}$$

Let the empirical loss function of the PINN be

$$L[f_k] := \frac{1}{M}\sum_j \frac{1}{N}\sum_i \int \left| \frac{\partial f_k}{\partial t}(x_j,t_i) - \mathcal{L}f_k(x_j,t_i) \right|^2 \mathrm{d}x\,\mathrm{d}t,$$

where (x_j) are uniform sample from our compact domain of interest and $(t_i)_{i=1}^N$ is the uniform grid for $[0,T]$.

Find the sequence of functions ϕ_k, $k \ge 1$, *independent* of the samples $(x_j), (t_i)$, such that f_k is a probability density function and $\partial_t f_k$ is continuous everywhere in t, and $\lim_{k\to\infty} L[f_k] = 0$, but $\lim_{k\to\infty} |f_k(x,t) - p(x,t)| \neq 0$ for any $t > 0$. What is the pointwise limit f of your f_k ?

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3 Project: Operator Learning

Consider the following high dimensional elliptic PDE of the committor function q [1]:

$$\begin{aligned} -\varepsilon\Delta q(\boldsymbol{x}) + \boldsymbol{b}(\boldsymbol{x})\cdot\nabla q(\boldsymbol{x}) = 0 \text{ in } \Omega\backslash(A\cup B),\\ q(\boldsymbol{x})|_{\partial A} = 0,\ q(\boldsymbol{x})|_{\partial B} = 1. \end{aligned} \tag{13}$$

where $\Omega\subset\mathbb{R}^d$ and A, B are two disjoint open set.

Let the drift term $\boldsymbol{b}(\boldsymbol{x}) = -\nabla V(\boldsymbol{x})$ with

$$V(\boldsymbol{x}) = 2\exp\big(-\|\boldsymbol{x}-\boldsymbol{a}\|^2/2\big)+2\exp\big(-\|\boldsymbol{x}-\boldsymbol{b}\|^2/2\big)+\exp\big(-\|\boldsymbol{x}-\boldsymbol{c}\|^2/2\big)+f(\boldsymbol{x})$$

where $\boldsymbol{a} = (-1,-1,\ldots,-1)$, $\boldsymbol{b} = (1,1,\ldots,1)$, $\boldsymbol{c} = (0,0,\ldots,0)$. The function f is regarded as the input of the system in the operator learning.

Set A as the (L^2)-balls centered at $\boldsymbol{a}$ with the radius 0.5 and B as the union of two balls centered at $\boldsymbol{b}$ and $\boldsymbol{b}$ with the same radius 0.5. Ω is set as a sufficiently large torus with periodic condition, say $[-10,10]^d$.

The task is to learn the map from the function f to the function q. We assume f is a periodic function defined on Ω within an L^2 ball $\|f\|_2/|\Omega|\le 0.1$.

The Fourier Neural Operator can be found at the website https://zongyi-li.github.io/

- Implement FNO to solve this problem at least $d\ge 10$. You may assume the input f is already in the form of Fourier expansion.
- How would you sample f?
- How do you make sure q is always between 0 and 1?
- How to measure the errors of your algorithm? How to validate your numerical result is true?
- How to improve the accuracy ?
- If the input includes both ε and f, how to incorporate this additional parameter ?

[1] Find the meaning of committor function in the Transition Path Theory

In Transition Path Theory: The meaning of committor function q(x)

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4 WKB methods/Laplace's method

4.1 1D

The mean exit time (MET) of a one-dimensional Ornstein-Uhlenbeck (OU) process from an interval $[-1, 1]$ is found by solving the boundary-value problem

$$\epsilon^2 u''(x) - xu(x) = -1$$

with $u(0) = u(1) = 0$. The WKB method assume the asymptotic expansion for small ϵ is

$$u(x) \sim e^{C(x)/\epsilon^2}(u_0(x) + \epsilon^\alpha u_1(x) + \cdots)$$

Find the value of α and the Eikonal equation for $C(x)$ and the transport equation for $u_0(x)$. Note the transient layer near the boundary.

4.2 Kramers formula

Hendrik Antony Kramers was the first to translate the question of determining the rate of a chemical reaction into properties of PDEs [2]. Consider a one-dimensional over-damped Langevin

$$\mathrm{d}X_t^\varepsilon = -V'(X^\varepsilon)\,\mathrm{d}t + \sqrt{2\varepsilon}B_t$$

Let a and b be the two only local (strictly) minimum points of V, separated by a (non-degenerate) saddle point c.

Prove the following Kramers formula for small ε:

$$\begin{aligned}&\mathbb{E}\left[\inf\left\{t > 0 : X_t^\varepsilon = b\right\} \mid X_0^\varepsilon = a\right] \\ &\quad = \left[1 + o(1)_{\varepsilon\to 0}\right] \frac{2\pi}{\sqrt{V''(a)\,|V''(c)|}} \exp\left\{\varepsilon^{-1}\left(V(c) - V(a)\right)\right\},\end{aligned}$$

Generalize this formula to dimension d.

[2] Hänggi, P., Talkner, P., Borkovec, M.: Reaction-rate theory: fifty years after Kramers. Rev. Mod. Phys. 62(2), 251–342 (1990).
Kramers, H.A.: Brownian motion in a field of force and the diffusion model of chemical reactions. Physica 7(4), 284–304 (1940)

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5 McKean-Vlaso example

Consider the following equation :

$$dX(t) = [-\alpha X(t) + \mathbb{E}(X(t))]dt + dW(t), \quad t \geq 0$$

where W is a standard Wiener process; α is a positive constant. $X_0 = x_0$ is given. The distribution of X_t is denoted by μ_t, which can be shown as Gaussian.

1. Discuss the invariant measures when $\alpha \neq 1$ and $\alpha = 1$.
2. Study the time limit of μ_t as $t \to +\infty$.
3. Write a code to simulate this mean-filed SDE to validate your findings.
4. Consider the N-particle interactive system $dX_i(t) = [-\alpha X_i(t) + \frac{1}{N}\sum_j X_j]dt + dW_i(t)$, $1 \leq i \leq N$.

   For each given N, what is the invariant measure of $\mathbf{X}^N(t) = (X_1(t), \dots X_N(t))$?
5. Write a code to validate if the propagation of chaos hold in this example.
6. Show that the empirical mean $\bar{X}^N(t) = \frac{1}{N}\sum_{i=1}^N X_i(t)$ is an OU process in the form of $\mathrm{d}\bar{X}^N(t) = b\bar{X}^N(t)\ \mathrm{d}t + \sigma_N\, \mathrm{d}\bar{W}_t$ where $\bar{W}$ is a standard Brown motion. Find the value of b and σ_N.
7. Let φ be a smooth test function with a compact support. Define $\langle \mu_t, \varphi \rangle = \int \varphi(x)\mu_t(dx) = \mathbb{E}\varphi(X_t)$. Show that $\frac{d}{dt}\langle \mu_t, \varphi \rangle = \langle \mu_t, L(\mu_t)\varphi \rangle$ where

   $$L(\mu_t)\phi(x) := \frac{1}{2}\Delta\phi(x) + (-\alpha x + \int x\mu_t(dx)) \cdot \nabla\phi(x)$$

   is the infinitesimal generator parametrized by μ_t.
8. Introduce the empirical measure $\mu_t^N(\cdot) = \frac{1}{N}\sum_i \delta_{X_i(t)}(\cdot)$ and define the random variable $\Phi^N(t) = \langle \mu_t^N, \Phi \rangle = \frac{1}{N}\sum_{i=1}^N \Phi(X_i(t))$ for a test function Φ.

   - Derive the SDE that Φ^N satisfies:

   $$\Phi^N(t) - \Phi^N(0) = \int_0^t \left\langle \mu_s^N, L(\mu_s^N)\Phi \right\rangle \mathrm{d}s + M_t^N \tag{14}$$

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where

$$M_t^N = \frac{1}{N}\sum_{i=1}^{N}\int_0^t \nabla\Phi(X_i(s))\,\mathrm{d}W_i(s)$$

is the Ito integral aggregates the contribution of all N Brownian motions (W^i). Thus, M_t^N is a martingale.

- Use Ito isometry to prove that

$$\mathbb{E}\left[\left(M_t^N\right)^2\right] = \mathbb{E}\int_0^t \frac{1}{N}\left\langle \mu_s^N, \|\nabla\Phi\|^2\right\rangle \mathrm{d}s = O\left(\frac{1}{N}\right) \tag{15}$$

- For a martingale, the natural "size" of an increment $\mathrm{d}M_t^N$ over time dt is $\sqrt{\mathrm{d}\langle M^N\rangle_t}$, where $\langle M^N\rangle_t$ is the quadratic variance of the martingale M_t^N. So that the generalized Ito isometry holds:

$$\mathbb{E}\left[\left(\int_0^t H_s\,\mathrm{d}M_s\right)^2\right] = \mathbb{E}\int_0^t H_s^2\,\mathrm{d}\langle M\rangle_s.$$

Write $M_t^N = \sum_i Y^i$ where $Y_i = \frac{1}{N}\int_0^t \nabla\Phi(X_i(s))\,\mathrm{d}W_i(s)$. Then by the bilinear of quadratic variance, $\langle M_t^N\rangle = \sum_i\sum_j\langle Y_i, Y_j\rangle_t$.
By the rule of quadratic variance

$$\left[\int HdX, \int KdZ\right]_t = \int_0^t H_sK_sd[X,Z]_s$$

show that $\langle Y_i, Y_j\rangle_t = \frac{1}{N^2}\int_0^t \nabla\Phi(X_i)\cdot\nabla\Phi(X_j)\,\mathrm{d}\langle W_i, W_j\rangle_s = \frac{1}{N^2}\int \nabla_0^t\Phi(X_i)\cdot\nabla\Phi(X_j)\delta_{ij}\,\mathrm{d}s$ and prove that

$$\mathrm{d}\langle M_t^N\rangle = \frac{1}{N}\left\langle \mu_t^N, \|\nabla\Phi\|^2\right\rangle$$

i.e. the pure noise term for the SDE of Φ^N (or μ^N) is $\mathrm{d}M_t^N \approx O(\frac{1}{\sqrt{N}})$

6 colored noise

This example involves a mean-field model for an active particle X subject to an active force modeled as colored noise. The dynamics is :

$$\begin{aligned}\mathrm{d}X^\epsilon &= \left(-X^\epsilon + \epsilon^{-1}\eta - \int_{\mathbb{R}}(X^\epsilon - X')p_\epsilon(X',t)\,\mathrm{d}X'\right)\mathrm{d}t,\\ \mathrm{d}\eta^\epsilon &= \left(-\epsilon^{-2}\eta^\epsilon\right)\ \mathrm{d}t + \sqrt{2\epsilon^{-2}}\,\mathrm{d}B_\eta.\end{aligned} \tag{16}$$

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B_η is a standard Brownian motion and p_ϵ is the pdf of X^ϵ. The Ornstein-Uhlenbeck (OU) process η^ϵ is used to model the colored noise with the stationary distribution $\mathcal{N}(0,1)$.

1. Show that for any positive ϵ, the stationary distribution of the original system (16) is unique and is the zero-mean Gaussian distribution with the following covariance matrix Σ_ϵ:

$$\Sigma_\epsilon = \begin{bmatrix} \frac{1}{2(1+2\epsilon^2)} & \frac{\epsilon}{1+2\epsilon^2} \\ \frac{\epsilon}{1+2\epsilon^2} & 1 \end{bmatrix}.$$

   which converges to $\begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{bmatrix}$ as $\varepsilon \to 0$.

2. Compute the (stationary) correlation function $\mathrm{Cov}(\epsilon^{-1}\eta_t, \epsilon^{-1}\eta_{t+s})$ for any two positive s, t when the process η is at equilibrium. What is the limit as $\epsilon \to 0$.

3. Let the infinitesimal generator of (16) be L_ϵ and compute its asymptotic expansion formally as $\epsilon \to 0$.

4. Define the Itô SDE driven by an independent Brownian motion denoted by B_x:

$$\mathrm{d}\bar{X} = \left(-\bar{X} - \int (\bar{X} - \bar{X}')p(\bar{X}', t)d\bar{X}'\right)\mathrm{d}t + \sqrt{2}\,\mathrm{d}B_x \tag{17}$$

   Show that the process X^ϵ weakly converges to the process $\bar{X}$.

5. Do you have strong convergence $\mathbb{E}\|X_t^\epsilon - \bar{X}_t\|$?

6. Numerically verify your above conclusions.

7 foundation: Year 1-2 概率与线性代数基础

1. Now let L be an n by n positive semi-definite matrix and define the $(n-1)$ by $(n-1)$ matrix , G, as follows : $G_{ij} = L_{nn}L_{ij} - L_{ni}L_{nj}$ for $i = 1, \ldots, n-1$. Show that G is also positive semi-definite.

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2. Given a random variable X_0 whose density function is denoted by p_0, define $X_t = \alpha_t X_0 + \sigma_t \epsilon$ where ϵ is the standard normal random variable. $t > 0$ is fixed and σ and α are two real numbers.

   (a) Write the formula for p_t, the density function of X_t.

$$p_t(x) = \int p_{t|0}(x \mid x_0)\, p_0(x_0)\, dx_0 = \int \frac{1}{(2\pi\sigma_t^2)^{d/2}} \exp\left(-\frac{\|x - \alpha_t x_0\|^2}{2\sigma_t^2}\right) p_0(x_0)\, dx_0$$

   (b) Derive the mathematical expression of the so-called conditional score function $s(x_0, x_t, t) = \nabla_{x_t} \log p_{t|0}(x_t \mid x_0) = \dfrac{\nabla_{x_t} p_{t|0}(x_t \mid x_0)}{p_{t|0}(x_t \mid x_0)}$ where

$$p_{t|0}(x_t \mid x_0) = \Pr(X_t = x_t \mid X_0 = x_0)$$

   which is the condition density function of X_t given X_0.

   (c) Using Bayes' theorem for conditional densities, derive the mathematical expression of the function $p_{0|t}(x_0 \mid x_t) := p(X_0 = x_0 | X_t = x_t)$ which is the condition density function of X_0 conditioned on $X_t = x_t$.

   (d) Prove that

$$\mathbb{E}[s(X_0, X_t, t) \mid X_t = x] = \nabla \log p_t(x) = \frac{\nabla p_t(x)}{p_t(x)} \tag{18}$$

   where $p_t(x)$ is the probability density function of X_t. We may call $\nabla \log p_t(x)$ the marginal score. (18) states that **the marginal score is the conditional expectation of the conditional score**.

   (e) To have the above equality, do you need assume the condition that $X_t = \alpha_t X_0 + \sigma_t \epsilon$? Why?

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8 Fokker-Planck equation and numerical discretization

Consider the Fokker-Planck equation in R^1

$$\partial_t p_t = \boldsymbol{\nabla} \cdot (p_t \nabla V) + \frac{1}{2}\Delta p_t$$

on a finite interval, say, $I = [0, 1]$. The initial condition is a given continuous pdf p_0. The boundary condition is the Neumann boundary condition: $\partial_x p_t = 0$ at two ends $x = 0$ and 1.

Given a mesh grid partition of the domain $I = \cup_{i=1}^n I_i$ where $I_i = (x_i - h/2, x_i + h/2)$ and the equal mesh size $h = |I|/n$ and the grid $x_i = (i - 1/2)h$. Use finite volume method or some finite difference method to derive the discrete system for

$$\rho_i(t) \approx p(t, x_i).$$

Then the numerical approximation

$$\rho(t, x) = \sum_i \rho_i(x)\chi_{I_i}(x)$$

Do not discrete the continuous time. Can your scheme make sure ρ_i always satisfy $\int_I \rho(t, x)dx = 1$ for $t > 0$ and $\rho_i \geq 0$. Your scheme should be in the form of a forward equation for a Markov Process with discrete state space $\{I_i\}_{i=1}^n$.

Let $b_i := -\nabla V(x_i)$.

1. The central difference method

$$\dot{\rho}_i = -\frac{\rho_{i+1}b_{i+1} - \rho_{i-1}b_{i-1}}{2h} + \frac{1}{2}\frac{\rho_{i+1} - 2\rho_i + \rho_{i-1}}{h^2}$$

with $i = 1, \ldots, n$ and $\rho_0 = \rho_1$ and $\rho_{n+1} = \rho_{n-1}$.

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9 Foundation: ODE flow and Probability flow

Consider an initial value problem of the ODE in $\mathbb{R}^d$:

$$\dot{x}_t = v_t(x_t), \quad t \in [s, +\infty) \tag{19}$$

with an initial at time s: $x_s = x$. Assume that $v : (t, x) \in \mathbb{R} \times \mathbb{R}^d \mapsto \mathbb{R}^d$ is smooth and $C_0(\mathbb{R}^d; \mathbb{R}^d)$ for each t. Define the flow map $\Phi_{s,t} : x_s \mapsto x_t$ with $t \geq s$, which is a nonlinear operator on $\mathbb{R}^d$. By convention and abuse of notation, define $\Phi_t = \Phi_{0,t}$ if v_t independent of t.

For convenience, denote $\boldsymbol{\nabla} \cdot v_t(x) = \mathrm{Tr}(Dv_t(x))$ as $\theta_t(x)$

1. If $v_t(x) = A(t)x + B(t)$ where $A(t) \in \mathbb{R}^{d\times d}$ and $B(t) \in \mathbb{R}^d$ are two known functions. Write the solution x_t for a given initial $x_s = x$.

2. Show that $\Phi_{s,t}$ satisfies the semigroup property: $\Phi_{t+h,t}\Phi_{t,s} = \Phi_{t+h,s}$ for any $s, t, h \geq 0$. Show the inverse $\Phi_t^{-1} = \Phi_{-t}$ for $t \geq 0$ and give the precise meaning of Φ_{-t}. Write the ODE for the time-reversed process defined by $y_t = x_{T-t}$ with $t \in [0, T]$.

3. Let $M_t(x) := D\Phi_t(x)$ be the d-by-d Jacobi matrix of Φ_t, which is the derivative of x_t with respect to the initial $x_0 = x$. Show that for each x, the dynamics of the Jacobi matrix reads $\dot{M}_t(x) = Dv_t(x_t)M_t(x)$ and $M_0(x) = I_d$ (identity matrix), which can be written as

$$D\Phi_t(x) = \exp\left(\int_0^t Dv_t(x_t)\, \mathrm{d}s\right), \quad \forall x$$

where $x_t = \Phi_t(x)$

hint: you may need the fact $\det\exp(A) = \exp(\mathrm{Tr}(A))$ *for a square matrix* A *or Jacobi's formula that* $\frac{d}{dt}\det A(t) = \det A(t)\,\mathrm{Tr}\left(A(t)^{-1}\frac{d}{dt}A(t)\right)$.

4. Define the Jabobian determinant $J_t(x) = \det(M_t(x))$ for each x, Show that

$$J_t(x) = \exp\left(\int_0^t \boldsymbol{\nabla} \cdot v_t(x_t)\, \mathrm{d}t\right), \quad \forall x. \tag{20}$$

where $x_t = \Phi_t(x)$.

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If the forward Euler scheme with constant step size $h = 1/K > 0$ is used to discretize the ODE(19) within the time interval $[0, 1]$:

$$x_{k+1} = V_k(x_k), \quad k \geq 0$$

by letting $V_k(x) := x + hv_k(x)$. Show $\log(\det DV_k(x)) = h\boldsymbol{\nabla} \cdot v_{t_k}(x) + O(h^2)$. What is the Jacobian determinant $\widehat{J}_K(x_0)$ of the (discrete-time) flow map $\widehat{\Phi}_K : x_0 \mapsto x_K$ with $K = 1/h$?

What is the order of convergence for the error between $\widehat{J}_K = \det\left(D\widehat{\Phi}_K\right)$ and $J_{t=1} = \det(D\Phi_{t=1})$ in terms of h? What if you use a better ODE solver? Validate this convergence order numerically using an illustrative example.

5. Let $\Omega_0 \subset \mathbb{R}^d$. Define $\Omega_t = \Phi_t(\Omega_0) = \{\Phi_t(x_0) : x_0 \in \Omega_0\}$, the deformed domain at time t under the flow of the ODE (19). Let $A_t := \text{vol}(\Omega_t)$, the Lebesgue volume. Is the below formula true ?

$$\dot{A}_t = \int_{\Omega_t} \boldsymbol{\nabla} \cdot v_t(x)\, \mathrm{d}x. \tag{21}$$

6. **Fix** two points y, z in $\mathbb{R}^d$, find the optimal vector field v_t^* such that $\Phi_1(y) = z$ and $I[v] := \int_0^1 \|v_t(\Phi_t(y))\|^2 w_t \mathrm{d}t$ is minimal, where $\|v_t\|$ is the L_2 norm. At this optimal v^*, write the flow map Φ_t as the function of $t \in [0, 1]$, y and z.

7. Let X_0 be a $\mathbb{R}^d$-valued random variable with the probability density function ρ_0. Then the distribution of the random variable $X_t := \Phi_t(X_0)$ is called the push-forward measure of ρ_0 under the map Φ_t, denoted by $\rho_t = (\Phi_t)_\# \rho_0$. Derive the continuity equation

$$\partial_t \rho_t(x) + \boldsymbol{\nabla} \cdot (v_t(x)\rho_t(x)) = 0$$

that ρ_t satisfies by using weak formulation with a test function $f \in C_C^\infty(\mathbb{R}^d)$ and the fact $\frac{d}{dt}\mathbb{E}f(X_t) = \mathbb{E}\frac{d}{dt}f(X_t)$.

8. Using change of variable to show that

$$\rho_t(x_t) = \frac{\rho_0(x_0)}{J_t(x_0)} = \rho_0(x_0) \exp\left(-\int_0^t \boldsymbol{\nabla} \cdot v_t(x_t)\right), \quad \forall t \geq 0 \tag{22}$$

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where $x_t = \Phi_t(x_0)$ and verify the continuity equation by using (20). Show that this implies the trival fact

$$\int f(x)\rho_t(x)\mathrm{d}x = \mathbb{E}f(X_t) = \mathbb{E}f(\Phi_t(X_0)) = \int f(\Phi_t(x))\rho_0(x)\mathrm{d}x \quad (23)$$

9. Let $F_t(x)$ be a smooth scalar-valued test function on $[0,T]\times\mathbb{R}^d$. Note $X_t = \Phi_t(X_0) \sim \rho_t$ and the density function ρ_t satisfies the continuity equation. Note $\int_{\mathbb{R}^d}\rho_t(x)F_t(x)dx = \mathbb{E}F_t(X_t)$. Show the following Transport Theorem by the chain rule of differentiation and the continuity equation:

$$\frac{\mathrm{d}}{\mathrm{d}t}\ (\mathbb{E}F_t(X_t)) = \int \rho_t(x)\frac{\mathrm{D}F_t}{\mathrm{D}t}(x)\ \mathrm{d}x = \mathbb{E}\frac{\mathrm{D}F_t}{\mathrm{D}t}(X_t) \quad (24)$$

where $\frac{\mathrm{D}}{\mathrm{D}t} = \partial_t + v_t\cdot\nabla$ is the total derivative along the flow. This derivative is also called the material derivative, the derivative following the flow.

Compare it with

$$\int F_t(x)\partial_t\rho(x)\mathrm{d}x = -\int F_t(x)\boldsymbol{\nabla}\cdot(v_t(x)\rho_t(x))\mathrm{d}x = \int \rho_t(x)v_t(x)\cdot\nabla F_t(x)\mathrm{d}x \quad (25)$$

For convenience, denote $\boldsymbol{\nabla}\cdot v_t(x) = \mathrm{Tr}(Dv_t(x))$ as $\theta_t(x)$

10. Apply this formula (23) or (25) to $F_t(x) = -\log\rho_t(x)$ to derive the evolution of the differential entropy

$$H(\rho_t) = -\int \rho_t(x)\log\rho_t(x)\ \mathrm{d}x = \mathbb{E}[-\log\rho_t(X_t)]$$

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}H(\rho_t) &= \int \rho_t(x)\boldsymbol{\nabla}\cdot(v_t(x))\ \mathrm{d}x = \int \rho_t(x)\theta_t(x)\mathrm{d}x\\ &= \mathbb{E}\theta_t(X_t) = \mathbb{E}\theta_t(\Phi_t(X_0))\\ &= \int \rho_0(x)\theta_t(\Phi_t(x)))\ \mathrm{d}x\end{aligned}$$

and explain the meaning of this formula.

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By using $\partial_t \rho_t + v_t \cdot \nabla \rho_t = -\rho_t \boldsymbol{\nabla} \cdot v_t$, show that

$$\frac{\mathrm{d}}{\mathrm{d}t} H(\rho_t) = -\int v_t(x) \cdot \nabla \log \rho_t(x)\ \rho_t(x) \mathrm{d}x = \mathbb{E}[-v_t(X_t) \cdot s_t(X_t)] \quad (26)$$

where the Stein score function

$$s_t(\cdot) := \nabla \log \rho_t(\cdot).$$

11. $s_t(X_t)$ appears in (26) is in the Lagrangian frame and can be written as the composition of the score function in Euclidean frame and the flow map: $G_t = s_t \circ \Phi_t : X_0 = x \mapsto X_t \mapsto \nabla \log \rho_t(X_t)$. Show for each x, the ODE for $G_t(x)$ holds:

$$\frac{\mathrm{d}}{\mathrm{d}t} s_t(X_t) = \frac{\mathrm{d}}{\mathrm{d}t} G_t(x) = -(Dv_t(X_t))^\top G_t(x) - \nabla(\boldsymbol{\nabla} \cdot v_t(X_t)),$$

with the initial $G_0(x) = \nabla \log \rho_0(x)$. Dv is the Jacobian matrix of v and $\boldsymbol{\nabla} \cdot v = \mathrm{tr}(Dv)$ is the divergence of v. Discuss the long time behavior as $t \to \infty$. Generalize this result to the $\frac{\mathrm{d}}{\mathrm{d}t} \nabla^2 \log \rho_t(X_t) = \frac{\mathrm{d}}{\mathrm{d}t} D s_t(X_t)$

12. Now consider two distributions ρ_t and ρ_t^* associated with v and v^* respectively. Define the relative entropy (Kullback–Leibler divergence) $D_{\mathrm{KL}(\rho_t \| \rho_t^*)} = \int \rho_t(x) \log \frac{\rho_t(x)}{\rho_t^*(x)}\ \mathrm{d}x$. Show that

$$\frac{\mathrm{d}}{\mathrm{d}t} D_{\mathrm{KL}}(\rho_t \| \rho_t^*) = -\int \rho_t(x)(v_t(x) - v_t^*(x)) \cdot (s_t(x) - s_t^*(x))\ \mathrm{d}x \quad (27)$$

where $s_t^*(x) = \nabla \log \rho_t^*(x)$ is the Stein score function of ρ_t^*.

13. Given two probability density functions q and ρ_0.

Let $J[\Phi_1] = \mathrm{KL}(q \| \rho_1) := \int q(x) \log \frac{q(x)}{\rho_1(x)} \mathrm{d}x = \int q \log q - \int q \log \rho_1$ where $\rho_1 = (\Phi_1)_{\#} \rho_0$ and Φ_1 is the flow map associated with the vector field v. Calculate the first variation J' of the functional $J[\Phi_1]$ w.r.t. the flow map Φ_1 i.e., $J[\Phi_1 + \delta\Phi_1] - J[\Phi_1] \approx \int J'(x_1) \cdot \delta\Phi_1(x_0)\ \mathrm{d}x_1$ with respect to an infinitesimal perturbation $\Phi_1 \to \Phi_1 + \delta\Phi_1$, where $x_1 = \Phi_1(x_0)$ or $x_0 = \Phi_1^{-1}(x_1)$.

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14. Let ρ be the standard Gaussian distribution in $\mathbb{R}^d$ and $h(x) \geq 0$ be a function such that $Z_h := \int h(x)\rho(x)\, dx < \infty$. Define a new distribution $\rho_h = h(x)\rho(x)/Z_h$. Construct a flow map such that if $x_0 \sim \rho$, then $x_1 = \Phi_t(x_0) \sim \rho_h$. What if you do not have the expression of ρ and only the samples of ρ, how to design the algorithm?

15. To minimze $\frac{1}{2}\int_0^1 \int \rho_t(x)\|v_t(x)\|^2 dxdt + E[\rho_1]$ where ρ_0 is given and $\rho_t = (\Phi_t)_{\#}\rho_0$ and $\Phi_t$is the flow map of the vector fieldv_t.

10 Hamiltonian flow

Consider a special case of (19), Hamiltonian flow where $x_t = (q_t, p_t)$ and $v(x) = (\nabla_p H(x), -\nabla_q H(x))$ for a given Hamiltonian function $H(x) = \frac{1}{2}|p|^2 + V(q)$.

1. Write the ODE for q_t and p_t.

2. Show that $H(x(t)) \equiv H(x(0))$ for all t.

3. Define the time reverse of x_t as $\tilde{x}_t := x_{T-t} = (q_{T-t}, p_{T-t})$. What is the ODE for $\tilde{x}_t$? Is it also Hamiltonian? How would you define the time reverse instead?

4. Show that the Hamiltonian flow map Φ_t is symplectic, i.e., $(D\Phi_t)^\top J D\Phi_t = J$ with $J = \begin{pmatrix} 0 & I_d \\ -I_d & 0 \end{pmatrix}$.

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11 Relative entropy(Kullback–Leibler divergence)

For a probability measure θ on a complete separate metric space $\mathcal{X}$, the relative entropy (Kullback–Leibler divergence), $R(\cdot\|\theta)$, is a mapping from a probability measure γ to an extended real number, defined by

$$R(\gamma\|\theta) := \int_{\mathcal{X}} \left(\log \frac{d\gamma}{d\theta}\right) d\gamma \tag{28}$$

whenever γ is absolutely continuous with respect to θ; otherwise, we set $R(\gamma\|\theta) = +\infty$.

1. (Jensen's Inequality) Prove that $R(\gamma\|\theta) \geq 0$, and the equality attains if and only if $\gamma = \theta$.

2. Find an example that $R(\gamma\|\theta) \neq R(\theta\|\gamma)$.

3. Let θ be the distribution of a standard Gaussian variable Z in $\mathbb{R}$ and γ_b is the conditional distribution of Z that $Z > b$. Plot $R(\gamma_b\|\theta)$ as a function of the real number b. Solve the following optimization problem:
$$\inf_{b\geq 0} \left\{ R(\gamma_b\|\theta) + \int U \, \mathrm{d}\gamma_b \right\}$$
where U is a quadratic function $U(x) = \lambda x^2/2$ for a positive parameter λ.

4. Show that $R(\gamma\|\theta)$ is convex in the pair of probability measures (γ, θ). Is R convex in γ ? Is R convex in θ? Bregmann divergence

5. (Varadhan's variational formula) Let U be a bounded measurable function such that $\int e^{-U} d\theta < \infty$. Define the exponentially tilted distribution γ_U by
$$\frac{\mathrm{d}\gamma_U}{\mathrm{d}\theta}(x) := \frac{e^{-U(x)}}{\int e^{-U(x')} \, \mathrm{d}\theta(x')}$$
Sometimes, it is also written as $\mathrm{d}\gamma_U(x) \propto e^{-U(x)} \, \mathrm{d}\theta(x)$.

   Show that
$$R(\gamma\|\theta) + \int U d\gamma = R(\gamma\|\gamma_U) - \log \int e^{-U} d\theta \geq -\log \int e^{-U} d\theta \tag{29}$$

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and consequently

$$\inf_{\gamma}\left\{R(\gamma\|\theta)+\int U d\gamma\right\}=-\log\int e^{-U}d\theta \tag{30}$$

with the infimum attained uniquely at $\gamma^*=\gamma_U$.

6. **Donsker-Varadhan variational representation of the KL divergence**. This is to enhance the understanding of (29).

   Let p and q be probability distributions defined on a space $\mathcal{X}$. The Kullback-Leibler (KL) divergence satisfies:

   $$D(p\parallel q)=\sup_{g\in\mathcal{G}}\left(\mathbb{E}_{X\sim p}[g(X)]-\log\mathbb{E}_{X\sim q}\left[\exp(g(X))\right]\right)$$

   where $\mathcal{G}$ denotes the set of all bounded, measurable functions $g:\mathcal{X}\to\mathbb{R}$. Show the optimal g^* is the log ratio $\log(p/q)$.

   Interpret this result as a natural consequence of the Fenchel-Legendre duality. Fix q. Consider the *cumulant generating functional* $\Psi(g)=\log\mathbb{E}_{X\sim q}[e^{g(X)}]$ for every bounded function g and the KL divergence $\Psi^*(p)=D(p\parallel q)$ a functional of p. Show that they are both convex and satisfy **Fenchel-Legendre** duality:

   $$\Psi^*(p) \quad =\sup_g\Big(\langle g,p\rangle-\Psi(g)\Big) \tag{31}$$

   $$\Psi(g) \quad =\sup_p\Big(\langle g,p\rangle-\Psi^*(p)\Big) \tag{32}$$

7. (chain rule) Write a random variable $X=(X_1,X_2)\in\mathcal{X}_1\times\mathcal{X}_2$ componentwisely. μ and ν are two probability distributions on $\mathcal{X}_1\times\mathcal{X}_2$ and can be regarded as the joint distribution of $X=(X_1,X_2)$ and $Y=(Y_1,Y_2)$, respectively. Let μ_1 and ν_1 be the marginal distribution of X_1 and Y_1, respectively, i.e., $\mu_1(\cdot)=\int_{\mathcal{X}_2}\mu(\cdot,dx_2)$. Denote by $\mu_{2|1}(x_1,\cdot)$ the conditional distribution of X_2 with a given $X_1=x_1$.

   Show that

   $$R(\mu\|\ \nu))=R(\mu_1\|\ \nu_1)+\mathbb{E}_{x_1\sim\mu_1}R(\mu_{2|1}(x_1,\cdot)\|\ \nu_{2|1}(x_1,\cdot)) \tag{33}$$

   As a consequence, if X_1 and X_2 are independent, Y_1 and Y_2 are independent, then

   $$R(\mu\|\nu)=R(\mu_1\|\nu_1)+R(\mu_2\|\nu_2)$$

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8. [Stein's Identity]

   - For a standard normal random variable X, show that

   $$\mathbb{E}[f(X)X] = \mathbb{E}[\nabla f(X)] \tag{34}$$

   - Define Stein Operator associated with a pdf p which decays at infinity:

   $$\mathbf{A}_p f(x) = \nabla f(x) + f(x)\nabla \log p(x).$$

   A simple integration-by-part trick shows that

   $$\mathbb{E}_{X\sim p}[\mathbf{A}_p f(X)] = 0 \tag{35}$$

   for any test function f. This means any expectation for a gradient can be computed withou the gradient, via the score function $s = \nabla \log p$.

9. Given two probability distributions p and q on a space $\mathcal{X}$, and a convex function $f : (0, \infty) \to \mathbb{R}$ with $f(1) = 0$, the *f-divergence* is defined as

$$D_f(p\|q) = \int_{\mathcal{X}} q(x)\, f\left(\frac{p(x)}{q(x)}\right) dx.$$

Identify the KL divergence is a special case of f-divergence by specifying the corresponding f.

Prove that f-divergence is jointly convex in p and q.

Show f-divergence admits a variational representation (Nguyen-Wainwright-Jordan bound):

$$D_f(p\|q) = \sup_{T\in\mathcal{T}} \Big(\mathbb{E}_{x\sim p}[T(x)] - \mathbb{E}_{x\sim q}[f^*(T(x))]\Big),$$

where $T : \mathcal{X} \to \mathbb{R}$ is any measurable function (the discriminator in GANs). f^* is the convex conjugate of f, defined by

$$f^*(u) = \sup_{t>0} \big(ut - f(t)\big).$$

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10. Apply to the KL divergence and show that KL divergence satisfies

$$D(p\|q) = 1 + \sup_{u\in\mathcal{T}} \Big(\mathbb{E}_{x\sim p}[u(x)] - \mathbb{E}_{x\sim q}[\exp(u(X))] \Big), \tag{36}$$

To directly prove this, using Fenchel-Legendre for $\exp(u) = \sup_{s>0}(su - s\log s + s)$ and Sion theorem to show

$$D(p\|q) = 1 + \inf_{g(\cdot)} \left\{ \sup_{u} \left\{ \mathbb{E}_p[u] - \mathbb{E}_q[e^g u] \right\} + \mathbb{E}_q[ge^g - e^g] \right\}.$$

if writing $s(x) = e^{g(x)}$ in terms of a new function g. Show the optimal $g^* = \log(p/q)$ is the log ratio.

Compare the space of Fenchel-Legendre used there and that used in Donsker-Varadhan variational representation (29).

11. This is to show (36), as a special case of convex duality in (Nguyen-Wainwright- Jordan) , can leads to Varadhan-Donsker (29) due to the special invariance to shift in (36).

We take the function $u(X)$ in (36) and shift it upward by an arbitrary scalar constant c, meaning we substitute $u(X) \leftarrow g(X) + c$:

$$D(p \parallel q) \geq 1 + \mathbb{E}_{X\sim p}[g(X) + c] - \mathbb{E}_{X\sim q}\left[\exp(g(X) + c)\right]$$

$$D(p \parallel q) \geq 1 + \mathbb{E}_{X\sim p}[g(X)] + \max_{c} \left(c - e^c \mathbb{E}_{X\sim q}\left[\exp(g(X))\right]\right)$$

Thus it is easy to prove that

$$D(p \parallel q) \geq \mathbb{E}_{X\sim p}[g(X)] - \log \mathbb{E}_{X\sim q}\left[\exp(g(X))\right]$$

yields the Donsker-Varadhan representation.

12. Now reinterpret the density ratio estimation problem as a binary classification problem with the logistic ratio. Show that

$$\min_g \mathbb{E}_p\left[\log\left(1 + e^{-g(X)}\right)\right] + \mathbb{E}_q\left[\log\left(1 + e^{g(X)}\right)\right],$$

has the same solution as $g^* = \log(p/q)$.

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13. In generative modelling, the inference may also need to steer to a reward. Let $\mu \mapsto \mathcal{R}(\nu)$ is a measure-dependent function with some regularity and is assumed concave. Given a distribution ρ with the density function p, the aim is to sample

$$\mu^* = \arg\max_{\mu}(\mu - \mathrm{KL}(\mu\|\pi)) \tag{37}$$

Show that μ^* satisfies the mean-field-type condition in the following exponential form $\mu^*(x) \propto e^{\phi(x;\mu^*)}p(x)$. Find ϕ.

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12 model reduction and free energy

μ is a given probability distribution on$\mathbb{R}^n$, for instance, $\mu(\mathrm{d}x) = \mathcal{Z}^{-1}e^{-\beta U}\mathrm{d}x$ with a given inverse temperature $\beta > 0$ and a known potential function U.$\mathcal{Z} := \int e^{-\beta U}\mathrm{d}x < +\infty$. Let

$$\xi : x \in \mathbb{R}^n \to z \in \mathbb{R}^m$$

where $m \leq n$, be a smooth map. Assume that the rank of the Jacobi matrix $D\xi$ is m. Define $\nu = \xi_{\#}\mu$, the push-forward image of measureμ of the mapξ. The conditional distribution of giving $\xi(x) = z$ for a fixed z is denoted by $\nu(\mathrm{d}x|z)$, which only supports on the level set (a sub - manifold of$\mathbb{R}^n$ with codimension m)

$$\Sigma(z) = \{x : \xi(x) = z\}.$$

$\nu(\mathrm{d}x|z)$ satisfies $\mathbb{E}[f(X)|\xi(X) = z] = \int_{\mathbb{R}^n} f(x)\nu(\mathrm{d}x|z)$ for any test function$f$, and the normalization condition $\int_{\Sigma(z)} \nu(\mathrm{d}x|z) \equiv 1$for all$z$. Specifically,$\nu(\mathrm{d}z)and\nu(\mathrm{d}x|z)$a re defined as follows that for any two test funct ions$f$and$g$,

$$\int_{\mathbb{R}^n} g(\xi(x))\ f(x)\ \mu(\mathrm{d}x) = \int_{\mathbb{R}^m} g(z)\left(\int_{\Sigma(z)} f(x)\nu(\mathrm{d}x|z)\right)\ \nu(\mathrm{d}z). \tag{38}$$

This means the simple fact of double expectation formula:

$$\mathbb{E}_X[g(\xi(X))f(X)] = \mathbb{E}_Z\Big[g(Z)\mathbb{E}_X[f(X)|X \in \Sigma(Z)]\Big]$$

wheeX $\sim \mu$and$Z = \xi(Z)$. For convenience, we introduce the delta measu r e on$\mathbb{R}^n$with the support only on the submanifold$\Sigma(z)$:

$$\int_{\mathbb{R}^n} g(\xi(x))\ f(x)\ \ \mathrm{d}x = \int_{\mathbb{R}^m} g(z)\left(\int_{\Sigma(z)} f(x)\delta_{\xi(x)-z}(\mathrm{d}x)\right)\ \mathrm{d}z. \tag{39}$$

1. Let μ be the standard Gaussian distribution in $\mathbb{R}^n$ and $\xi(x) = \frac{1}{n}(x_1 + \ldots + x_n)$. What is the distribution of ν and $\nu(\mathrm{d}x|z)$? How to understand the delta distribution $\delta_{\xi(x)-z}(\mathrm{d}x)$.

2. Define a random variable $Z_\epsilon = Z + \varepsilon = \xi(X) + \varepsilon$ where ε is the $\mathbb{R}^m$-valued Gaussian random variable with mean zero and standard deviation $\epsilon > 0$. ε and X are independent. Then the conditional distribution $\nu_\epsilon(\mathrm{d}x|z)$ can be defined likewisely. Based on the limit $\lim_\epsilon \nu_\epsilon(\mathrm{d}x|z) = \nu(\mathrm{d}x|z)$, derive an approximation expression for the delta function which is abstractly defined by (39).

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3. To derive the density function for ν and $\nu(\cdot|z)$ w.r.t. the Lebesgue measure, the following co-arear formula is central:

$$\int_{\mathbb{R}^n} f(x)\sqrt{\det G(x)}\mathrm{d}x = \int_{\mathbb{R}^m} \widetilde{f}(z)\,\mathrm{d}z$$

where $\widetilde{f}(z) = \int_{\Sigma(z)} f(x)\,\mathrm{d}\sigma_{\Sigma(z)}(x)$ is the integral on the surface $\Sigma(z)$ with a given z. $\mathrm{d}\sigma_{\Sigma(z)}(x)$ is the Lebesque measure on $\Sigma(z)$.

$$G(x) = (D\xi)(D\xi)^\top$$

is the $m \times m$ Gram matrix. This result is a generalization of the change-of-variable if ξ is one-to-one and $m = n$.

Show that if $\mu(\mathrm{d}x) = \mu(x)\mathrm{d}x = \mathcal{Z}^{-1}e^{-\beta U}\mathrm{d}x$, then $\nu(\mathrm{d}z) = \mathbb{P}(\xi(X) = z)\,\mathrm{d}z$ can be written as $\nu(\mathrm{d}z) = e^{-\beta F(z)}\,\mathrm{d}z$ where F is called the *free energy*, taking the form

$$\begin{aligned} F(z) &= -\beta^{-1}\log\left(\int_{\Sigma(z)} \mu(x)\ \delta_{\xi(x)-z}(\mathrm{d}x)\right) \\ &= -\beta^{-1}\log\left(\int_{\Sigma(z)} \mu(x)\det(G(x))^{-1/2}\,\mathrm{d}\sigma_{\Sigma(z)}(x)\right) \end{aligned} \tag{40}$$

The conditional density takes the form

$$\nu(\mathrm{d}x|z) = e^{-\beta U(x)}(\det(G)(x)^{-1/2})\sigma_{\Sigma(z)}(\mathrm{d}x)/\int_{\Sigma(z)} e^{-\beta U(x)(\det(G)(x)^{-1/2})}\,\mathrm{d}\sigma_{\Sigma(z)}(x)$$

4. Show the gradient of the free energy, called mean force[3],

$$\nabla F(z) = \int_{\Sigma(z)} \vec{f}(x)\nu(\mathrm{d}x|z) = \mathbb{E}(\vec{f}(X)|\xi(X) = z)$$

where the $\mathbb{R}^m$-valued $\vec{f}(x)$ has the expression

$$f_i(x) = \vec{\ell}_i(x)\cdot\nabla U(x) - \beta^{-1}\boldsymbol{\nabla}\cdot\vec{\ell}_i(x) \tag{41}$$

where the vector $\vec{\ell}_i(x) = \sum_{j=1}^m (G^{-1})_{i,j}(x)\nabla\xi_j(x)$ is determined by the Jacobi matrix.

[3]score function in machine learning

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13 Effective Dynamics and Collective Variable

Consider the over-damped Langevin dynamics

$$\mathrm{d}\boldsymbol{X}_t = -\nabla V(\boldsymbol{X}_t)\,\mathrm{d}t + \sqrt{2\beta^{-1}}\,\mathrm{d}W_t \tag{42}$$

This dynamics is ergodic with respect to

$$\mu(\mathrm{d}\boldsymbol{x}) = Z^{-1}\exp(-\beta V)\,\mathrm{d}\boldsymbol{x}$$

Let the collective variable (reaction coordinate) be a smooth function

$$\xi : \mathbb{R}^d \to \mathbb{R}$$

which may describe the effective dynamics of the microscopic variables$\boldsymbol{X}$.

1. Show that

$$\mathrm{Var}[\boldsymbol{X}] = \mathrm{Var}\left[\mathbb{E}(\boldsymbol{X}|\xi(\boldsymbol{X}))\right] + \mathbb{E}[\mathrm{Var}(\boldsymbol{X}|\xi(\boldsymbol{X})]$$

   - As an illustration, let $\boldsymbol{X}$ follows a centered Gaussian distribution with the covariance matrix Σ. Assume ξ is a linear map $\xi(\boldsymbol{X}) = \boldsymbol{v}_\xi \cdot \boldsymbol{X}$. Starting with the above variance decomposition formula, what is the optimal choice of the vector $\boldsymbol{v}_\xi$ in order to heuristically approximate the distribution of $\boldsymbol{X}$?
   - To go beyond the Gaussian model, assume $V(\boldsymbol{x}) = U(x_1) + x_2^2/2$ where $U(x) = (x^2-1)^2/4$ is the double-well potential. What would be the best linear projection for $\xi(\boldsymbol{X}) = \boldsymbol{v}_\xi \cdot \boldsymbol{X}$?

2. Assume the function ξ is given, use the Ito formula to show that

$$d\xi\left(\boldsymbol{X}_t\right) = \left(-\nabla V \cdot \nabla\xi + \beta^{-1}\Delta\xi\right)\left(\boldsymbol{X}_t\right)dt + \sqrt{2\beta^{-1}}\left|\nabla\xi\left(\boldsymbol{X}_t\right)\right|\frac{\nabla\xi\left(\boldsymbol{X}_t\right)}{\left|\nabla\xi\left(\boldsymbol{X}_t\right)\right|}\cdot dW_t$$

3. Define the following SDE of $\boldsymbol{Z}_t$

$$\mathrm{d}\boldsymbol{Z}_t = b(t,\boldsymbol{Z}_t)\,\mathrm{d}t + \sqrt{2\beta^{-1}}\sigma(t,\boldsymbol{Z}_t)\frac{\nabla\xi\left(\boldsymbol{X}_t\right)}{\left|\nabla\xi\left(\boldsymbol{X}_t\right)\right|}\cdot dW_t \tag{43}$$

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where

$$b(t,z) = \mathbb{E}\left[\left(-\nabla V \cdot \nabla \xi + \beta^{-1}\Delta\xi\right)\big|\xi(\boldsymbol{X}_t) = z\right]$$
$$\sigma^2(t,z) = \mathbb{E}\left[|\nabla\xi(\boldsymbol{X}_t)|^2\big|\xi(\boldsymbol{X}_t) = z\right]$$

with $\boldsymbol{Z}_0 \sim \xi(\boldsymbol{X}_0)$. Show that $\boldsymbol{z}_t$ and $\xi(\boldsymbol{X}_t)$ share the same distribution for all t.

4. To do

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14 Stochastic Interpolants

[4] Define the following interpolation with $t \in [0,1]$, between two points as a linear map $\mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}^d$:

$$I_t(x_0, x_1) = \alpha_t x_0 + \beta_t x_1 \tag{44}$$

where $\alpha_0 = 1 = \beta_1$ and $\alpha_1 = 0 = \beta_0$. Let (X_0, X_1) satisfy the joint distribution $\pi(x_0, x_1)$ with the marginals ρ_0 and ρ_1, consider the random variable

$$X_t := I_t(X_0, X_1)$$

1. Assume X_0 and X_1 are two independent standard normal distribution in $\mathbb{R}^d$. Compute the distribution X_t and the covariance between X_t and X_s.

2. Let ρ_t be the pdf of X_t. Then $\rho_t = (I_t)_{\#}\pi$ and our goal is to model this time-dependent pdf. For example, ρ_t satisfies a continuity equation with the initial condition:

$$\partial_t \rho_t + \nabla \cdot (b_t \rho_t) = 0, \qquad \rho_{t=0} = \rho_0 \tag{45}$$

   How many choices of these b_t ? Describe these drift functions all.

   We can take a test function $A(x)$ and consider

$$\begin{aligned} \mathbb{E}A(X_t) &= \mathbb{E}A(I_t(X_0, X_1) \\ \int \rho_t(x) A(x)\,\mathrm{d}x &= \int\int A(I_t(x_0, x_1))\pi(x_0, x_1)\,\mathrm{d}x_0\,\mathrm{d}x_1 \end{aligned}$$

   Take the time derivatives, use the sifting property of the Dirac delta function and the integration by part, then

$$\begin{aligned} \int \partial_t \rho_t(x) A(x)\,\mathrm{d}x &= \iint \nabla A(I_t(x_0, x_1)) \cdot \dot{I}_t(x_0, x_1)\pi(x_0, x_1)\,\mathrm{d}x_0\,\mathrm{d}x_1 \\ &= \iint \left(\int \nabla A(x) \delta(x - I_t(x_0, x_1)\,\mathrm{d}x \right) \cdot \dot{I}_t(x_0, x_1)\pi(x_0, x_1)\,\mathrm{d}x_0\,\mathrm{d}x_1 \\ &= \int \nabla A(x) \cdot (v_t(x)\rho_t(x))\,\mathrm{d}x \\ &= -\int A(x)\ \operatorname{div}(v_t(x)\rho_t(x))\,\mathrm{d}x \end{aligned}$$

[4] https://jmlr.org/papers/v26/23-1605.html

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where

$$\begin{aligned} v_t(x) :=& \frac{1}{\rho_t(x)} \iint \delta(x - I_t(x_0, x_1)\dot{I}_t(x_0, x_1)\pi(x_0, x_1)\,\mathrm{d}x_0\,\mathrm{d}x_1 \\ =& \frac{1}{\mathbb{E}[\delta(X_t - x)]}\mathbb{E}\left[\delta(x - X_t)\dot{I}_t(X_0, X_1))\right] \\ =& \mathbb{E}\left[\dot{I}_t(X_0, X_1))\middle| X_t = x\right] \end{aligned} \tag{46}$$

by noting $E[A(X,Y) \mid B(X,Y) = b] = \frac{\iint A(x,y)\pi(x,y)\delta(B(x,y)-b)dxdy}{\iint \pi(x,y)\delta(B(x,y)-b)dxdy}$ for any two random variables X, Y with the joint π.

3. The score function $s_t(x) := \nabla \log \rho_t(x)$ satisfies $s_t(x) = \gamma(t)\mathbb{E}[X_0|X_t = x]$. Find γ.

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15 Feymann Kac formula

Consider the time-homogeneous Ito process satisfying

$$\mathrm{d}X_t = b(X_t)\,\mathrm{d}t + \sigma(X_t)\,\mathrm{d}W_t \tag{47}$$

The infinitesimal generator A is given by

$$Af = b \cdot \nabla f + \frac{1}{2}\,\mathrm{trace}(a\nabla^2 f) \tag{48}$$

where $a = \sigma\sigma^\top$.

1. The Feymann-Kac formula is that $u(x,t) := \mathbb{E}[f(X_t)|X_0 = x]$ satisfies the backward Kolmogorov equation: $\partial_t u = Au$ with $u(x,0) = f(x)$. Prove this conclusion by applying Ito lemma to $f(X_t)$.

2. The adjoint of A is the Fokker-Planck operator $Lp = -\nabla \cdot (bp) + \frac{1}{2}\,\mathrm{trace}(\nabla\nabla^\top(ap))$. Let p_t is the unique pdf of the Fokker-Planck equation and p_∞ is the unique invariant pdf. Both are strictly positive everywhere. Then define

   $$h(t,x) = p(t,x)/p_\infty(x)$$

   Show that h satisfies a new *backward* Kolmogorov equation: $\partial_t h = \widetilde{A}h$

3. Then prove the result about Schrodinger's bridge

   $$p(t,\mathbf{x}) = \mathbb{E}\left[\frac{p_0\left(X_t\right)}{p_\infty\left(X_t\right)}\middle|\, X_0 = \mathbf{x}\right] p_\infty(\mathbf{x})$$

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16 Jarzynski equality

Let $(\pi_t)_{0\le t\le T}$ be a curve of probability distribution. We write each π_t as a Boltzmann-Gibbs distribution

$$\pi_t = Z_t^{-1} e^{-\beta E_t(x)}$$

where E_t is the potential and the constant β is the inverse temperature. $Z_t = \int e^{-\beta E_t(x)} \mathrm{d}x$ is the normalization constant. We may construct a stochastic dynamics, say the over-damped Langevin dynamics from $t = 0$ to $t = T$:

$$\mathrm{d}X_t = -\nabla E_t(X_t)\ \mathrm{d}t + \sqrt{2/\beta}\,\mathrm{d}W_t \tag{49}$$

where $X_0 \sim \pi_0$. Note that the law of X_t is NOT equal to π_t for $t > 0$.

1. Show that $\int \mathcal{L}_t(\varphi)\,\mathrm{d}\pi_t(x) = 0, \forall t$ for any test function φ, where $\mathcal{L}_t$ is the infinitesimal generator of the above over-damped Langevin dynamics.

2. In non-equilibrium physics, X_t refers to the macroscopic slow variable. How to interpret the π_t as the "stationary" distribution in certain sense.

3. For each trajectory X, Define $A_t(\{X_s\}_{0\le s\le t}) = \int_0^t \ell(X_s)\,\mathrm{d}s$ where ℓ is a function of x variable. Consider the expected value under the weight $e^{-\beta A_t}$:

$$u(s,x) := \mathbb{E}\left[\varphi(X_t) e^{-\beta(A_t - A_s)} \mid X_s = x\right] \tag{50}$$

   Show that u satisfies the following backward Kolmogorov equation

$$\partial_s u + \mathcal{L}_s u = \beta \lambda_s'(x) u(s,x), 0 \le s \le t \tag{51}$$

   with the terminal condition $u(t,\cdot) = \varphi(\cdot)$. λ' is the time patial derivative of λ. Find the expression λ_s. Here are the hints.

   (a) Let $U(r, X_r) = u(r, X_r) \exp\left(-\beta \int_s^r \ell(X_\tau)\,\mathrm{d}\tau\right)$ for $s \le r \le t$, with $X_s = x$. Apply the Ito calculus to $\mathrm{d}U(r, X_t)$.

   (b) Show that if u satisfies (51), then $U(r, X_r)$ is a martingale for $r \in [s,t]$. Then $\mathbb{E}U(s, X_s) = \mathbb{E}U(t, X_t)$ which means that $u(s,x) = \mathbb{E}\left[u(t, X_t) \exp\left(-\beta \int_s^t \ell(X_r)\,\mathrm{d}r\right) | X_s = x\right] = \mathbb{E}\left[\varphi(X_t) \exp\left(-\beta \int_s^t \ell(X_r)\,\mathrm{d}r\right) | X_s = x\right]$

4. Using (51) to find the condition for ℓ_s and π_s such that

$$\frac{d}{ds}\left(\int u(s,x) e^{-\beta\lambda_s(x)} \mathrm{d}x\right) = 0$$

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5. Show that for each t and test function φ

$$\frac{Z_t}{Z_0}\mathbb{E}_{Y\sim\pi_t}\varphi(Y) = \mathbb{E}\left[\varphi(X_t)\exp\left(-\beta A_t(\{X_s\}_{0\leq s\leq t})\right)\right] \tag{52}$$

6. Can we generalize to time-dependent test function φ_t?

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17 Diffusion bridge and Doob's h transform

[5]Consider the over-damped Langevin dynamics

$$\mathrm{d}X_t = -\nabla U(X_t)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}W_t \tag{53}$$

with the initial condition $X_0 = x_0$. The pdf of X_t is denoted by p_t.

1. Let B be a subset of $\mathbb{R}^d$. For any $t \in [0,T]$, and x, define the conditional probability

$$h_t(x) = \mathbb{P}(X_T \in B | X_t = x) \tag{54}$$

Then (Dynkin's formula) h satisfies the backward Fokker Planck equation and the terminal condition $h_T(x) = 1_B(x)$:

$$\partial_t h + \mathcal{L}^* h = \partial_t h - \nabla U(x)\cdot\nabla h(x) + \Delta h = 0. \tag{55}$$

where $\mathcal{L}^*$ is the infinitesimal generator

2. Define the pdf

$$q_t(x) = \mathbb{P}(X_t = x | X_T \in B, X_0 = x_0)$$

for $0 \le t \le T$.

Show that

$$\begin{aligned} q_t(x) &= \frac{\mathbb{P}(X_t = x, X_T \in B | X_0 = x_0)}{\mathbb{P}(X_T \in B | X_0 = x_0)} \\ &= \frac{\mathbb{P}(X_T \in B | X_t = x, X_0 = x_0)}{\mathbb{P}(X_T \in B | X_0 = x_0)} \mathbb{P}(X_t = x | X_0 = x_0) \\ &= \frac{h_t(x)}{h_0(x)} p_t(x) \end{aligned}$$

and satisfies that $q_0(x) = \delta(x - x_0) = p_0(x)$, $q_T(x) = 1_B(x) = h_T(x)$ Derive the partial differential equation of this pdf q_t and find the corresponding SDE of Y_t whose pdf is q_t.

[5](reference book: Stochastic Differential Equations by Bernt Oksendal). Reference: V.De Bortoli, J.Thornton, J.Heng, and A.Doucet, "Diffusion Schrödinger bridge with applications to score-based generative modeling," Adv. Neural Inf. Process. Syst.34, 17695–17709 (2021),

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3. The Girsanov Theorem. Let Y_t solves

$$\mathrm{d}Y_t = -\nabla U(Y_t)\,\mathrm{d}t + v_t(Y_t)\;\mathrm{d}t + \sqrt{2}\,\mathrm{d}W_t \tag{56}$$

with the additional (time-dependent) drift term $v_t(\cdot)$.

Show that for any constant $T > 0$, for any bounded functional on the path space $C([0,T];\mathbb{R}^d)$, we have

$$\mathbb{E}[\varphi(X)] = \mathbb{E}\left[\varphi(Y)\exp\left(-\int_0^T \frac{v_t(Y_t)}{\sqrt{2}}\,\mathrm{d}W_t - \int_0^T \frac{\|v_t(Y_t)\|^2}{2}\,\mathrm{d}t\right)\right] \tag{57}$$

You may start with the discrete Euler-Maruyama scheme for X and Y with a small time step Δt and then take the limit $\Delta t \to 0$.

18 Wasserstein Gradient flow

By JKO scheme[6], Fokker-Planck equation can be viewed as the Wasserstein gradient flow of the free energy with entropy contribution

$$\mathcal{F}(\rho) = \int V\rho dx + \int \rho\log\rho dx \tag{58}$$

with the discrete scheme

$$\rho_{n+1} = \arg\min_{\rho}(\mathcal{F}(\rho) + \frac{1}{2\tau}W_2^2(\rho,\rho_n))$$

, where τ is the time stepsize. As $\tau \to 0$, ρ_n converges to the Wasserstein gradient flow (Fokker-Planck equatio)

$$\partial_t\rho = -\operatorname{grad}\mathcal{F}(\rho) = \nabla\cdot(\nabla V\rho + \rho\nabla\log\rho)$$

Now consider the family of the distribution in the parametric form $\rho(\theta)$ and we can construct the discrete scheme for the parameter θ:

$$\theta_{n+1} = \arg\min_{\theta}(\mathcal{F}(\rho(\theta_n)) + \frac{1}{\tau}W_2^2(\rho(\theta),\rho(\theta_n))).$$

What is the continuous-time dynamics $\partial_t\theta(t) =?$ for θ?

[6]Jordan, R., Kinderlehrer, D., Otto, F.: The variational formulation of the Fokker-Planck equation. SIAM J. Math. Anal. 29(1), 1–17 (1998)

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Now consider a specific family of the Gaussian mixture distributions with the weights $\{w_k\}$ and the centers $\{z_k\}$, sharing the same variance. $k = 1, \dots K$. Specify the above gradient flow for these parameters $\{w_k\}$ and $\{z_k\}$ corresponding to the Wasserstein gradient flow. Discuss the corresponding particle dynamics for these z_k with the dynamical weight.